



Configuring a particle filter

- To implement a particle filter to solve the indoor-navigation problem, we must be able to:
 1. Compute the conditional pdfs: $f(z_k | x_k^{(i)})$ and $f(x_k^{(i)} | x_{k-1}^{(i)})$, and
 2. Draw samples from the importance density $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$.
- This lesson presents the strategy that I chose in order to do so.
- First, recall the state equation for the vehicle:

$$\begin{bmatrix} \xi_{k+1} \\ \eta_{k+1} \end{bmatrix} = \begin{bmatrix} \xi_k \\ \eta_k \end{bmatrix} + \Delta t \begin{bmatrix} s_k \cos \theta_k \\ s_k \sin \theta_k \end{bmatrix},$$

where $s_k^m = s_k + w_k^s$ and $\theta_k^m = \theta_k + w_k^\theta$, and $w_k^s \sim \mathcal{N}(0, \sigma_s^2)$ and $w_k^\theta \sim \mathcal{N}(0, \sigma_\theta^2)$.

- The measurement equation modeled power loss (in dB) between Tx and Rx as:

$$\text{Loss} = (\text{loss at 1 m}) + 20 \log_{10}(d) + N_w \times (\text{loss per wall}) + S,$$

where random shadowing is modeled via $S \sim \mathcal{N}(0, \sigma_{ln}^2)$.



Computing $f(z_k | x_k^{(i)})$

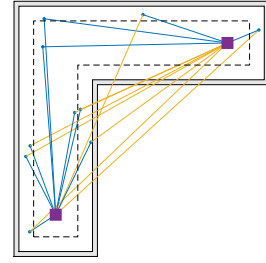
- The measurement z_k is a vector of power losses between the vehicle and each of the transmitting beacons.
- The deterministic portion of the power loss is computed as:

$$\bar{S} = (\text{loss at 1 m}) + 20 \log_{10}(d) + N_w \times (\text{loss per wall}),$$

where $d^2 = (\xi_k^{(i)} - x)^2 + (\eta_k^{(i)} - y)^2$ for vehicle location $(\xi_k^{(i)}, \eta_k^{(i)})$ and a beacon having coordinates (x, y) .

- Geometry is used to determine the number of walls traversed between $(\xi_k^{(i)}, \eta_k^{(i)})$ and (x, y) : blue paths for no walls and yellow paths for two walls.
- Since the random part of power loss $S \sim \mathcal{N}(0, \sigma_{ln}^2)$ is Gaussian and additive, the pdf we seek is also Gaussian:

$$f(z_k | x_k^{(i)}) = \mathcal{N}(\bar{S}, \sigma_{ln}^2).$$



A challenge with computing $f(x_k^{(i)} | x_{k-1}^{(i)})$

- It is much more challenging to find $f(x_k^{(i)} | x_{k-1}^{(i)})$.
- Let's consider only the horizontal position component for now:

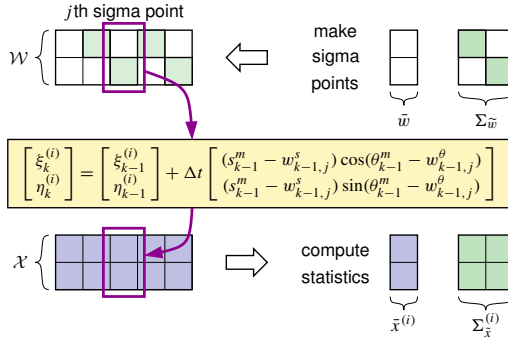
$$\begin{aligned} \xi_{k+1} &= \xi_k + \Delta t (s_k \cos \theta_k) = \xi_k + \Delta t \left((s_k^m - w_k^s) \cos(\theta_k^m - w_k^\theta) \right) \\ &= \xi_k + \Delta t \left((s_k^m - w_k^s) (\cos(\theta_k^m) \cos(w_k^\theta) + \sin(\theta_k^m) \sin(w_k^\theta)) \right) \\ &= \xi_k + \Delta t \left(s_k^m (\cos(\theta_k^m) \cos(w_k^\theta) + \sin(\theta_k^m) \sin(w_k^\theta)) \right) \\ &\quad - \Delta t \left(w_k^s (\cos(\theta_k^m) \cos(w_k^\theta) + \sin(\theta_k^m) \sin(w_k^\theta)) \right). \end{aligned}$$

- We know that $w_k^s \sim \mathcal{N}(0, \sigma_s^2)$ and $w_k^\theta \sim \mathcal{N}(0, \sigma_\theta^2)$, but how do we compute the distribution of $f(\xi_k^{(i)} | \xi_{k-1}^{(i)})$ using this information?
- It may be possible to derive a closed-form expression, but I chose to approximate it instead.



Method 1: Use sigma points

- I will assume that $f(x_k^{(i)} | x_{k-1}^{(i)})$ is Gaussian with some mean $\bar{x}^{(i)}$ and covariance $\Sigma_{\bar{x}}^{(i)}$: $f(x_k^{(i)} | x_{k-1}^{(i)}) = \mathcal{N}(\bar{x}^{(i)}, \Sigma_{\bar{x}}^{(i)})$.
- You learned in Lesson 3.2.2 that we can approximate the statistics at the output of a nonlinear function given the statistics of the input to the function using sigma points.



- Sigma points corresponding to w_k^s and w_k^θ are stored in $\mathcal{W}$.
- Compute sigma points $\mathcal{X}_j = f(\mathcal{W}_j)$.
- Compute output mean, covariance:

$$\bar{x}^{(i)} = \sum_{j=0}^p \alpha_j^{(m)} \mathcal{X}_j$$

$$\Sigma_{\bar{x}}^{(i)} = \sum_{j=0}^p \alpha_j^{(c)} (\mathcal{X}_j - \bar{x}^{(i)}) (\mathcal{X}_j - \bar{x}^{(i)})^T.$$



Method 2: Use a coordinate transformation

- I still assume that $f(x_k^{(i)} | x_{k-1}^{(i)}) = \mathcal{N}(\bar{x}^{(i)}, \Sigma_{\bar{x}}^{(i)})$.
- You learned in Lesson 2.4.2 that we can approximate noisy polar coordinates with noisy cartesian coordinates, which implies that:

$$\bar{x}^{(i)} \approx x_{k-1}^{(i)} + \Delta t \begin{bmatrix} s_m \cos \theta_m (1 - e^{-\sigma_\theta^2} + e^{-\sigma_\theta^2/2}) \\ s_m \sin \theta_m (1 - e^{-\sigma_\theta^2} + e^{-\sigma_\theta^2/2}) \end{bmatrix}, \text{ and } \Sigma_{\bar{x},k}^{(i)} \approx \begin{bmatrix} \Sigma_{\bar{x}}^{11} & \Sigma_{\bar{x}}^{12} \\ \Sigma_{\bar{x}}^{12} & \Sigma_{\bar{x}}^{22} \end{bmatrix}, \text{ where}$$

$$\begin{aligned} \Sigma_{\bar{x}}^{11} &= s_m^2 e^{-2\sigma_\theta^2} [\cos^2 \theta_m (\cosh 2\sigma_\theta^2 - \cosh \sigma_\theta^2) + \sin^2 \theta_m (2 \sinh 2\sigma_\theta^2 - \sinh \sigma_\theta^2) \\ &\quad + \sin^2 \theta_m (\sinh 2\sigma_\theta^2 - \sinh \sigma_\theta^2)] + \sigma_s^2 e^{-2\sigma_\theta^2} [\cos^2 \theta_m (2 \cosh 2\sigma_\theta^2 - \cosh \sigma_\theta^2)] \\ \Sigma_{\bar{x}}^{22} &= s_m^2 e^{-2\sigma_\theta^2} [\sin^2 \theta_m (\cosh 2\sigma_\theta^2 - \cosh \sigma_\theta^2) + \cos^2 \theta_m (\sinh 2\sigma_\theta^2 - \sinh \sigma_\theta^2)] \\ &\quad + \sigma_s^2 e^{-2\sigma_\theta^2} [\sin^2 \theta_m (2 \cosh 2\sigma_\theta^2 - \cosh \sigma_\theta^2) + \cos^2 \theta_m (2 \sinh 2\sigma_\theta^2 - \sinh \sigma_\theta^2)] \\ \Sigma_{\bar{x}}^{12} &= \sin \theta_m \cos \theta_m e^{-4\sigma_\theta^2} [\sigma_s^2 + (r_m^2 + \sigma_s^2)(1 - e^{\sigma_\theta^2})] = \Sigma_{\bar{x}}^{21}. \end{aligned}$$



Choosing an importance density $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$

- You have now learned how to compute $f(z_k | x_k^{(i)})$ and two ways to approximate $f(x_k^{(i)} | x_{k-1}^{(i)})$.
- We still need to choose an importance density $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$ from which we draw random samples to compute a new particle value $x_k^{(i)} \sim q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$.
- Then, we calculate the unnormalized weight for this particle:

$$\tilde{w}_k^{(i)} = \frac{f(z_k | x_k^{(i)}) f(x_k^{(i)} | x_{k-1}^{(i)})}{q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)} w_{k-1}^{(i)}.$$

- Since we have approximated $f(x_k^{(i)} | x_{k-1}^{(i)}) \approx \mathcal{N}(\bar{x}^{(i)}, \Sigma_{\bar{x}}^{(i)})$ and we can draw random Gaussian samples, I chose $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k) = f(x_k^{(i)} | x_{k-1}^{(i)})$.
- This simplifies the weight update equation to be: $\tilde{w}_k^{(i)} = f(z_k | x_k^{(i)}) w_{k-1}^{(i)}$.



Summary

- To implement a particle filter to solve the indoor-navigation problem, we must be able to:
 1. Compute the conditional pdfs: $f(z_k | x_k^{(i)})$ and $f(x_k^{(i)} | x_{k-1}^{(i)})$, and
 2. Draw samples from the importance density $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$.
- You learned how to compute $f(z_k | x_k^{(i)})$ for the path-loss model that I presented in the prior lesson.
- You learned that it may not be possible to compute $f(x_k^{(i)} | x_{k-1}^{(i)})$ exactly in a closed form, and two methods to approximate its value if we assume that the density is Gaussian.
 - A sigma-point approach and a coordinate-transformation approach.
- Finally, you learned the choice that I made for $q(x_k^{(i)} | x_{k-1}^{(i)}, z_k)$.
- Now, we're ready to implement the particle filter and see how it performs!